

NIPAWIN, SK, Canada

WMO: 713120

Lat: 53.3331N

Lon: 104.0047W

Elev: 372

StdP: 96.94

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.0	(c) -32.3	(d) -39.2	(e) 0.1	(f) -34.9	(g) -36.6	(h) 0.1	(i) -32.1	(j) 11.3	(k) -10.2	(l) 10.0	(m) -10.9	(n) 2.5	(o) 190	(p) 0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.7	(c) 28.7	(d) 18.9	(e) 26.9	(f) 18.1	(g) 25.3	(h) 17.3	(i) 20.9	(j) 26.4	(k) 19.6	(l) 24.9	(m) 18.5	(n) 23.5	(o) 4.7	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 18.9	(b) 14.4	(c) 23.8	(d) 17.6	(e) 13.2	(f) 22.3	(g) 16.5	(h) 12.3	(i) 21.1	(j) 62.3	(k) 26.6	(l) 57.8	(m) 24.8	(n) 54.1	(o) 23.7	(p) 27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.3	(b) 9.1	(c) 8.1	(e) -39.8	(f) 32.7	(g) 3.1	(h) 1.9	(i) -42.0	(j) 34.1	(k) -43.8	(l) 35.3	(m) -45.5	(n) 36.4	(o) -47.7	(p) 37.8		
(a) 10.3	(b) 9.1	(c) 8.1	(e) -39.9	(f) 23.5	(g) 3.1	(h) 1.3	(i) -42.1	(j) 24.4	(k) -44.0	(l) 25.2	(m) -45.7	(n) 25.9	(o) -48.0	(p) 26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.1	-17.7	-15.6	-8.6	2.1	9.9	15.3	17.6	16.7	11.5	3.4	-6.9	-15.0
	DBStd	14.16	8.88	7.87	7.77	6.21	4.95	3.57	2.79	3.23	4.41	4.82	6.79	8.18
	HDD10.0	3992	859	718	576	244	64	4	0	1	34	211	506	775
	HDD18.3	6333	1117	951	834	488	262	103	47	71	207	463	756	1033
	CDD10.0	759	0	0	0	6	62	162	237	208	79	6	0	0
	CDD18.3	60	0	0	0	0	2	11	25	19	2	0	0	0
	CDH23.3	857	0	0	0	3	65	153	271	284	79	2	0	0
	CDH26.7	185	0	0	0	1	14	32	47	73	18	0	0	0
	WSAvg	3.9	3.8	3.7	3.8	4.2	4.6	4.0	3.4	3.5	4.0	4.0	3.9	3.7
	PrecAvg	420	18	15	17	30	36	84	88	51	36	29	19	19
(15) Precipitation	PrecMax	575	40	27	58	84	93	169	162	123	85	81	43	43
	PrecMin	305	7	4	1	5	0	36	17	15	5	0	6	7
	PrecStd	78	9	6	14	19	24	34	44	29	26	21	10	10
	PrecStd	78	9	6	14	19	24	34	44	29	26	21	10	10
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	3.5	8.4	22.2	28.4	30.2	30.3	32.2	29.3	21.3	9.9	4.0
		MCWB	1.6	1.1	4.2	11.4	15.0	18.8	21.3	19.9	17.5	13.0	5.8	1.6
	2%	DB	0.8	0.7	5.1	18.6	24.9	26.9	27.9	28.7	25.5	16.8	6.3	1.0
		MCWB	-0.6	-0.6	2.3	9.5	13.2	17.6	20.2	18.7	15.7	9.9	3.4	-0.4
	5%	DB	-2.0	-1.7	3.4	14.9	22.6	24.9	26.3	26.5	22.7	13.5	3.7	-1.8
		MCWB	-3.0	-2.7	1.4	7.9	12.1	16.2	19.3	18.2	14.3	8.4	1.8	-2.8
	10%	DB	-4.7	-4.0	1.8	11.5	20.2	23.0	24.7	24.4	20.0	10.8	1.7	-3.8
		MCWB	-5.4	-5.0	0.3	6.2	11.0	15.2	18.4	17.0	13.3	7.2	0.3	-4.5
	0.4%	WB	1.7	1.5	4.5	12.2	16.5	21.2	23.5	21.6	18.6	13.5	6.2	1.9
		MCDB	3.4	3.1	8.1	21.0	24.2	27.1	28.1	28.0	25.7	20.3	9.2	3.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-0.5	-0.7	2.6	10.2	14.6	19.1	21.5	20.1	16.8	10.8	3.7	-0.4
		MCDB	0.7	0.6	4.7	17.0	22.7	24.6	26.4	26.4	24.3	14.8	5.9	0.9
	5%	WB	-2.9	-2.7	1.5	8.4	13.1	17.7	20.2	19.0	15.2	9.0	1.9	-2.8
		MCDB	-2.0	-1.8	3.2	14.3	20.4	23.0	25.0	25.0	21.3	12.6	3.6	-1.9
	10%	WB	-5.3	-5.1	0.4	6.5	11.8	16.4	19.1	17.9	13.6	7.5	0.4	-4.6
		MCDB	-4.5	-4.2	1.8	11.3	19.0	21.2	23.6	23.0	19.1	10.6	1.7	-3.9
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	10.4	10.8	11.3	13.6	11.7	11.7	12.7	12.4	9.7	8.1	8.8
		MCDBR	11.1	10.1	10.1	17.3	17.9	15.3	14.5	16.7	17.8	15.2	10.4	10.0
		MCWBR	10.1	9.2	7.9	9.8	8.5	7.6	8.3	8.3	9.2	9.3	7.9	9.0
	5% WB	MCDBR	10.9	10.5	9.6	15.8	15.4	12.9	13.4	14.4	16.3	14.0	9.8	9.8
		MCWBR	9.9	9.2	7.6	9.5	7.9	7.2	8.5	8.0	8.9	8.8	7.6	8.9
(40) Clear-Sky Solar Irradiance	taub		0.247	0.255	0.281	0.329	0.344	0.373	0.373	0.368	0.316	0.287	0.247	0.248
	taud		2.274	2.305	2.335	2.393	2.407	2.363	2.370	2.389	2.520	2.550	2.469	2.238
	Ebn at Noon		749	867	913	905	905	877	869	853	868	829	768	662
	Edh at Noon		59	79	96	104	109	115	113	104	81	63	49	50
	RadAvg		0.99	2.08	3.64	4.90	5.65	5.66	5.85	4.86	3.40	1.93	0.90	0.68
(44) All-Sky Solar Radiation	RadStd		0.09	0.12	0.25	0.41	0.51	0.42	0.37	0.41	0.34	0.30	0.12	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page